

Problem

Solve the problem in Example 13.1 (see Fig. 13.9) using the T-equivalent model for the magnetically coupled coils.

Example 13.1:

Calculate the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.9.

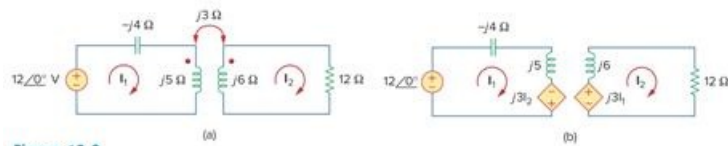


Figure 13.9
For Example 13.1.

Solution:

For loop 1, KVL gives

$$-12 + (-j4 + j5)\mathbf{I}_1 - j3\mathbf{I}_2 = 0$$

or

$$j\mathbf{I}_1 - j3\mathbf{I}_2 = 12 \quad (13.1.1)$$

For loop 2, KVL gives

$$-j3\mathbf{I}_1 + (12 + j6)\mathbf{I}_2 = 0$$

or

$$\mathbf{I}_1 = \frac{(12 + j6)\mathbf{I}_2}{j3} = (2 - j4)\mathbf{I}_2 \quad (13.1.2)$$

Substituting this in Eq. (13.1.1), we get

$$(j2 + 4 - j3)\mathbf{I}_2 = (4 - j)\mathbf{I}_2 = 12$$

or

$$\mathbf{I}_2 = \frac{12}{4 - j} = 2.91 \angle 14.04^\circ \text{ A} \quad (13.1.3)$$

From Eqs. (13.1.2) and (13.1.3),

$$\begin{aligned} \mathbf{I}_1 &= (2 - j4)\mathbf{I}_2 = (4.472 \angle -63.43^\circ)(2.91 \angle 14.04^\circ) \\ &= 13.01 \angle -49.39^\circ \text{ A} \end{aligned}$$

Step-by-step solution

Step 1 of 8

Refer to the Figure 13.9 in the textbook.

We need to replace the magnetically coupled coils with the T equivalent circuit. The relevant portion of the circuit is

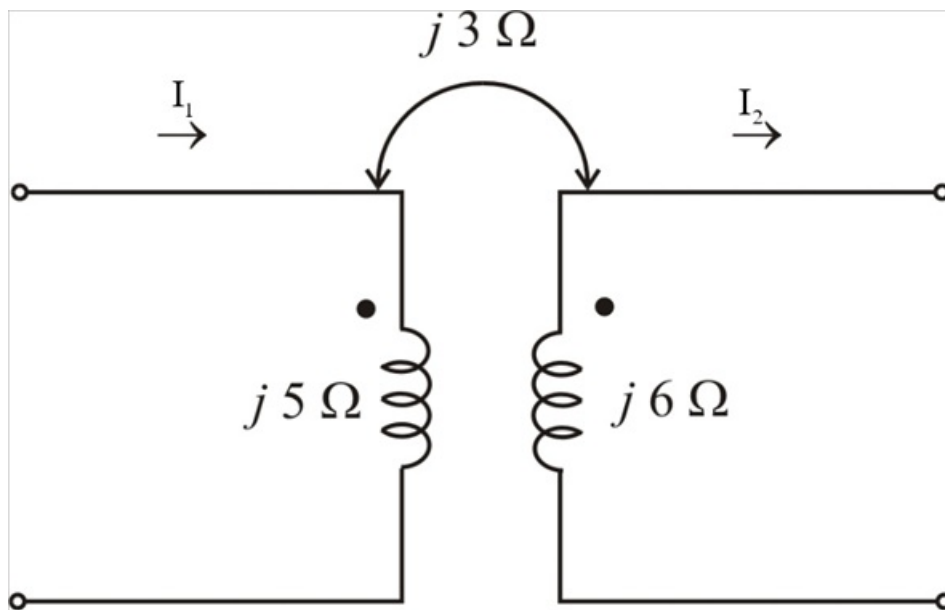


Figure 1

Step 2 of 8

Given that $L_1 = 5$, $L_2 = 6$ and $M = 3$, the T-equivalent network has the parameters as

Assume $\omega = 1$ rad/sec

$$L_a = L_1 - (-M)$$

$$L_a = 5 - (-3)$$

$$\therefore L_a = 8 \text{ H}$$

$$L_b = L_2 - (-M)$$

$$L_b = 6 - (-3)$$

$$\therefore L_b = 9 \text{ H}$$

$$L_c = -M$$

$$\therefore L_c = -3 \text{ H}$$

Step 3 of 8

The T-equivalent circuit is

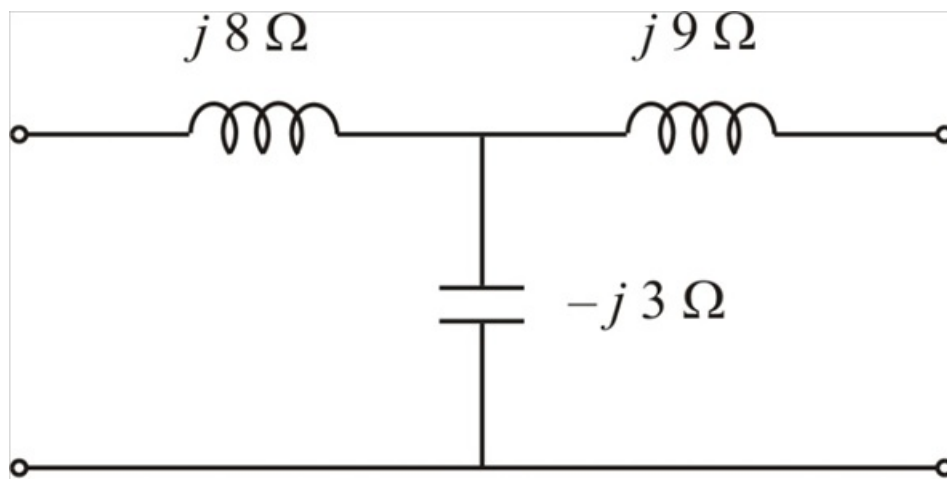


Figure 2

Step 4 of 8

Therefore, the equivalent circuit for the given circuit is

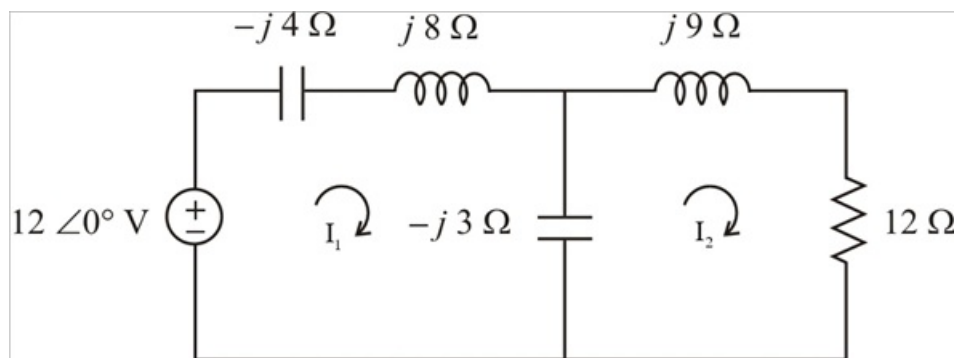


Figure 3

Step 5 of 8

Apply mesh analysis in mesh 1,

$$-12\angle 0^\circ - j4I_1 + j8I_1 - j3(I_1 - I_2) = 0$$

$$(-j4 + j8 - j3)I_1 + j3I_2 = 12\angle 0^\circ$$

$$jI_1 + j3I_2 = 12\angle 0^\circ$$

$$I_1 + 3I_2 = 12\angle -90^\circ$$

Rewrite the above equation.

$$I_1 = 12\angle -90^\circ - 3I_2 \dots\dots (1)$$

Step 6 of 8

Apply mesh analysis in mesh 2,

$$-j3(I_2 - I_1) + j9I_2 + 12I_2 = 0$$

$$j3I_1 + (j6 + 12)I_2 = 0$$

Step 7 of 8

Substitute the value of I_1 .

$$(j3)(12\angle -90^\circ - 3I_2) + (12 + j6)I_2 = 0$$

$$36 + (-j9 + 12 + j6)I_2 = 0$$

$$I_2 = \frac{-36}{12 - j3}$$

$$= 2.91\angle -14.04^\circ$$

Hence, the value of the mesh current I_2 is $2.91\angle -14.04^\circ \text{ A}$.

[Comments \(1\)](#)



Anonymous

value should be positive 14.04

Step 8 of 8

Substitute $2.91\angle -14.04^\circ$ for I_2 in equation (1).

$$I_1 = 12\angle -90^\circ - 3(2.91\angle -14.04^\circ)$$

$$= (12\angle -90^\circ) - (8.73\angle -14.04^\circ)$$

$$= -8.47 - j9.88$$

$$= 13\angle -49.4^\circ$$

Hence, the value of the mesh current is I_2 $13\angle -49.4^\circ \text{ A}$.